

DNANet Family One-Page Experiment Overview

NUDT-SIRST

IRSTD-1K

Key Takeaways

- 1. NUDT-SIRST: LDEM and LDEM-Gate both outperform DNANet.
- 2. NUDT-SIRST best performer: DNANet-LDEM (0.8409 best mIoU).
- 3. NUDT-SIRST most stable final result: DNANet-LDEM-Gate (0.8395).
- 4. IRSTD-1K best peak: DNANet-LDEM-Gate (0.6522 best mIoU).
- 5. IRSTD-1K final drop: DNANet-LDEM-Gate falls to 0.5232.
- 6. V2 solves peak performance, but not late-stage stability.

V3 Direction

Gate stabilization + semantic counter-correction to reduce late-stage oscillation on IRSTD-1K.

	Model	Best mIoU	Best PD	Best FA	Final mIoU	Gap
NUDT-SIRST	DNANet	0.8118	0.9619	1.59e-05	0.8114	0.0004
	DNANet-LDEM	0.8409	0.9757	1.58e-05	0.8218	0.0191
	DNANet-LDEM-Gate	0.8397	0.9725	1.05e-05	0.8395	0.0002

	Model	Best mIoU	Best PD	Best FA	Final mIoU	Gap
IRSTD-1K	DNANet	0.6446	0.9259	2.30e-05	0.6282	0.0163
	DNANet-LDEM	0.6499	0.8620	3.33e-05	0.6373	0.0125
	DNANet-LDEM-Gate	0.6522	0.9293	3.36e-05	0.5232	0.1290

